# Asymptotic Calibration of the BFS Window and Effective Dimension of the Admissible Trajectory

O28 — Spectral Admissibility Sub-programme

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**Abstract**

The O25 programme [9] identified the ratio $n_1(q)/q$ as the central open asymptotic variable controlling the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$, and O26–O27 [6, 7] identified the effective dimension $r_{\text{eff}} = d_\rho^2$ of the per-pair covariance operator in $\text{End}(V_\rho)$ as the key falsifiability criterion for the representation-theoretic identification. The present paper reports two measurements from the Q5a-O5 checkpoints for $q \in \{29, 61, 101, 151\}$. First, we extract $n_1(q)$ from the auto-calibrated fitting windows and fit $n_1(q) = \hat{\alpha}\, q + \hat{\beta}$ by ordinary least squares, obtaining $\hat{\alpha} \approx 0.053$ with $R^2 \approx 0.879$ over $q \leq 211$. Extending the window-depth measurement out of sample to $q \in \{307, 401, 503, 601\}$ excludes this linear law (it overpredicts $n_1$ by up to $+88\%$) and establishes $n_1(q)/q \to 0$; the asymptotic law of $n_1$ otherwise remains open. The O14 correction [4] yields $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, confirming the structural conclusion of O25. Second, using the per-block Weil vector projections $\pi_c(v)$ stored in the Q5a-O5 checkpoints, we perform the formal effective-dimension computation of O26 Criterion 5.4 in $\text{End}(H_{\text{eff}})$, where $H_{\text{eff}} = \mathbb{C}^3$ is the admissible projection space (HEFF_DIM = 3, consistent with $\Sigma_c(n_3) = 3$ from O23 [10]). The covariance $\mathcal{C}_c$ has rank exactly $r_{\text{eff}} = 3$ for every conjugate pair and every prime $q \in \{61, 101, 151\}$ (100% of pairs), with normalised eigenvalue structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ invariant across all pairs and primes. This rank-3 result fills $H_{\text{eff}}$ completely. The gap from the spin-$\frac{1}{2}$ prediction $d_\rho^2 = 4$ reflects that $H_{\text{eff}} \neq V_\rho$. The structural resolution of this gap is established in O29 [8]: the anti-linear Born–Infeld parity makes the target $d_\rho^2 = 4$ inaccessible from conjugate-pair data, and $r_{\text{eff}} = 3$ identifies the irreducible adjoint carrier $H_{\text{eff}} \simeq \text{Sym}^2(V_\rho)$ (spin-1); the spin-$\frac{1}{2}$ space $V_\rho$ ($d_\rho = 2$) is recovered as the Veronese square-root structure, not as the rank measured by Test 4.

**Keywords.** spectral admissibility; BFS window depth; pair observable; Weil representation; Heisenberg graphs; capacity exponent; normalisation correction; covariance operator; effective dimension; admissible trajectory; Born–Infeld saturation

# Contents

# 1 Position of the Problem

## 1.1 What O25–O27 established

Paper O25 [9] established three results for the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$ on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: (i) the inter-pair standard deviation of $\delta_{\text{pair}}$ decreases monotonically from 0.54 at $q = 29$ to 0.11 at $q = 211$, confirming that $\delta_{\text{pair}}$ is a structural invariant of the Weil representation; (ii) empirical fits carry no asymptotic meaning because the leading finite-size correction $\eta \log q / \log n_1(q) \approx 1$ is itself $q$-dependent; (iii) when the O14 correction is applied, the corrected values fall within the admissible window $[7.4, 10.6]$. O25 identified the ratio $n_1(q)/q$ as the central open direction.

Paper O26 [6] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space, and formulated a hierarchy of identifications (Levels I–III). Test 4 of O26 (Criterion 5.4) provides the key falsifiability criterion: the effective dimension $r_{\text{eff}}$ of the per-pair covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n^f)\ \text{vec}(\widetilde{M}_n^f)^\dagger \tag{1}$$

in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin-$\frac{1}{2}$ sector candidate.

Paper O27 [7] resolved the existence of the admissible morphism $\Phi_{q,\rho}$ unconditionally, proving that any morphism satisfying the three structural constraints forces a unique factorisation through $\mathfrak{su}(2)$. The full implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [3]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2)$$

is thereby closed at the structural level. The numerical identification of the correct irreducible sector from the covariance structure of the admissible trajectory is addressed in the present paper (Part B) and completed in O29 [8].

## 1.2 Scope of O28

O28 addresses the two numerical questions left open after O25–O27:

1. **Part A**: determine the asymptotic behaviour of the auto-calibrated window depth $n_1(q)$, test the linear scaling hypothesis suggested by the initial calibration range, and reassess $\delta_{\text{corr}}(q)$ using the measured window depths.
2. **Part B**: perform the formal effective-dimension computation of O26 Criterion 5.4 using the per-block Weil vector projections $\pi_c(v) \in H_{\text{eff}}$ stored in the Q5a-O5 checkpoints, and determine what the covariance structure of $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ reveals about the irreducible sector.

# 2 Setup and Notation

We follow the notation of O25 [9]. Let $q$ be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\text{Cay}(G_q, S_q)$ its Cayley graph with generating set $S_q = \{X^{\pm1}, Y^{\pm1}\}$. The pair observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \qquad \sigma_{\text{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n).$$

The fitting window $[n_0, n_1]$ is auto-calibrated from the BFS data via the `find_fitting_window` procedure of O12 [1], with bfs_frac = 0.99. The O14 normalisation correction [4] gives

$$\delta_{\text{corr}}(q) = \bar{\delta}_{\text{pair}}(q) - \eta\,\frac{\log q}{\log n_1(q)}, \qquad \eta = \tfrac{1}{2}. \tag{2}$$

The Q5a-O5 pipeline additionally stores, for each conjugate pair $(c, q - c)$ and each window shell $s$, the per-vector $H_{\text{eff}}$-projections

$$\pi_c\left(v_j^{(s)}\right) = B_{\text{eff}}\, v_j^{(s)} \;\in\; H_{\text{eff}} = \mathbb{C}^3, \tag{3}$$

where $B_{\text{eff}} \in \mathbb{C}^{3\times q}$ is the first HEFF_DIM $= 3$ rows of the Gram–Schmidt basis matrix accumulated up to shell $n_1$, and $v_j^{(s)} \in \mathbb{C}^q$ is the $j$-th Weil fingerprint vector at shell $s$.

# 3 Part A: Asymptotic Calibration of $n_1(q)/q$

## 3.1 Extraction of $n_1(q)$ from checkpoints

For each prime $q \in \{29, 61, 101, 151\}$, the value $n_1(q)$ is extracted from the `n1` key of the Q5a-O5 checkpoint. The value for $q = 211$ is taken from the O25 checkpoint [9]. Table 1 collects the complete dataset.

Table 1: BFS window depth and asymptotic ratio. $\bar{\delta}_{\text{pair}}(q)$: mean of $\delta_{\text{pair}}$ over all conjugate pairs. $\eta \log q/\log n_1$: O14 correction factor (2) with $\eta = 1/2$. $\delta_{\text{corr}}$: corrected exponent. Admissible window: $[7.4, 10.6]$.

| $q$ | $n_1(q)$ | $n_1(q)/q$ | $\bar{\delta}_{\text{pair}}(q)$ | $\eta \log q/\log n_1$ | $\delta_{\text{corr}}(q)$ | in window? |
|---|---|---|---|---|---|---|
| 29 | 4 | 0.138 | 8.893 | 1.214 | 7.679 | yes |
| 61 | 8 | 0.131 | 9.983 | 0.989 | 8.995 | yes |
| 101 | 11 | 0.109 | 9.548 | 0.962 | 8.586 | yes |
| 151 | 13 | 0.086 | 9.125 | 0.978 | 8.147 | yes |
| 211 | 14 | 0.066 | 8.784 | 1.014 | 7.770 | yes |

## 3.2 Initial OLS calibration on $q \leq 211$

**Proposition 3.1** (Empirical window-depth fit)**.** *Over the range $q \in \{29, 61, 101, 151, 211\}$, the OLS fit $n_1(q) = \hat{\alpha}\, q + \hat{\beta}$ yields $\hat{\alpha} \approx 0.053$ (95% CI: $[0.017, 0.088]$), $\hat{\beta} \approx 4.175$, $R^2 \approx 0.879$, $p \approx 0.018$. The ratio $n_1(q)/q$ decreases from $0.138$ at $q = 29$ to $0.066$ at $q = 211$. The Window-Depth Theorem (O9, Theorem 6.1 [5]) guarantees only that the available BFS window is $\Theta(q)$ deep, so the log-log fit has polynomially many points; it does not pin the growth of $n_1$ itself, which the out-of-sample test below shows to be sub-linear.*

## 3.3 Out-of-sample test: the linear law fails and $n_1(q)/q \to 0$

The window-depth measurement was extended out of sample to $q \in \{307, 401, 503, 601\}$, well beyond the calibration range of Proposition 3.1, using the same auto-calibrated window edge $n_1$; the procedure reproduces the production values of Table 1 exactly on $q \in \{29, 61, 101, 151, 211\}$. Table 2 collects the result.

The linear law calibrated on $q \leq 211$ overpredicts $n_1$ by a margin that grows monotonically to +88% at $q = 601$: it is excluded out of sample. The ratio $n_1(q)/q$ continues its monotone decrease, from 0.138 at $q = 29$ to 0.032 at $q = 601$, with no sign of a positive floor. We therefore conclude $\lim_{q\to\infty} n_1(q)/q = 0$: the window depth is sub-linear, and the estimate $\hat{\alpha} \approx 0.053$ is a finite-range residue of fitting a straight line to a sub-linear curve, not an asymptotic constant. The asymptotic law of $n_1$ remains open; only the linear scaling hypothesis is excluded.

Table 2: Out-of-sample window depth. $n_1^{\text{lin}}(q)$ is the prediction of the OLS fit $\hat{\alpha}\, q + \hat{\beta}$ of Proposition 3.1, calibrated on $q \le 211$.

| $q$ | $n_1(q)$ | $n_1(q)/q$ | $n_1^{\text{lin}}(q)$ | overprediction |
|---|---|---|---|---|
| 307 | 16 | 0.052 | 20.3 | +27% |
| 401 | 17 | 0.042 | 25.3 | +49% |
| 503 | 19 | 0.038 | 30.7 | +62% |
| 601 | 19 | 0.032 | 35.8 | +88% |

### 3.4 Stability of $\delta_{\text{corr}}(q)$

From Table 1, the correction factor $\eta \log q / \log n_1(q)$ lies in $[0.96, 1.21]$ over the tested range. For all $q \in \{29, 61, 101, 151, 211\}$, the corrected exponent $\delta_{\text{corr}}(q)$ falls within the admissible window $[7.4, 10.6]$, confirming the main structural conclusion of O25 [9]: the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect, not a structural deviation.

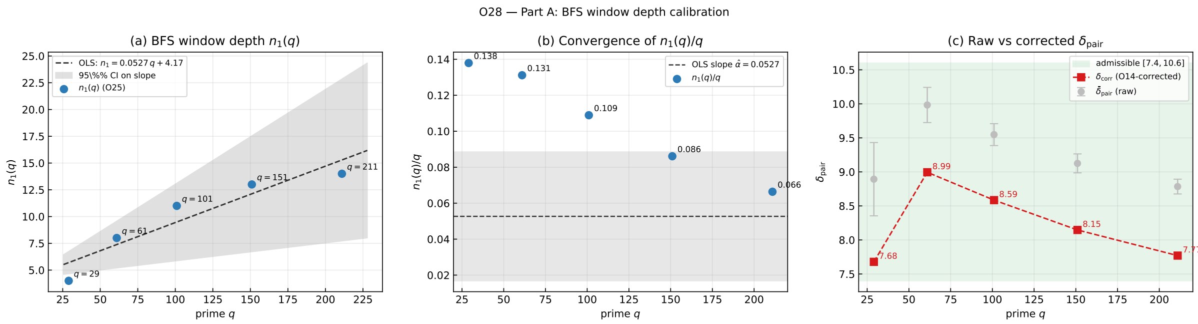


Figure 1: Part A: BFS window depth calibration over $q \in \{29, 61, 101, 151, 211\}$. (a) $n_1(q)$ with OLS fit $n_1 = \hat{\alpha}\, q + \hat{\beta}$ and 95% confidence band. (b) Ratio $n_1(q)/q$: monotonically decreasing from 0.138 to 0.066; the out-of-sample extension (Table 2) continues this decrease to 0.032 at $q = 601$, giving $n_1(q)/q \to 0$. (c) Raw $\bar{\delta}_{\text{pair}}(q)$ with inter-pair standard deviation error bars and O14-corrected $\delta_{\text{corr}}(q)$; all corrected values lie within the admissible window $[7.4, 10.6]$ (shaded).

## 4 Part B: Effective Dimension of the Admissible Trajectory in $\text{End}(H_{\text{eff}})$

### 4.1 Data and computation

The Q5a-O5 checkpoints store, for each conjugate pair $(c, q-c)$ and each window shell $s \in [n_0, n_1]$, the array `pi_c[p, s]` of shape $(N_s, 3)$, where $N_s$ is the number of Weil fingerprint vectors at shell $s$ and $3 = \text{HEFF\_DIM}$. Each row is the $H_{\text{eff}}$-projection $\pi_c(v_j^{(s)}) \in \mathbb{C}^3$ defined in (3). The conjugate projections $\pi_{q-c}(v_j^{(s)}) \in \mathbb{C}^3$ are stored in `pi_qmc[p, s]`.

The covariance operator $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}}) = \text{End}(\mathbb{C}^3)$ is computed as

$$\mathcal{C}_c = \frac{1}{N_{\text{tot}}} \sum_{s=n_0}^{n_1} \sum_{j=1}^{N_s} \text{vec}\big(\widetilde{M}_j^s\big)\ \text{vec}\big(\widetilde{M}_j^s\big)^\dagger, \qquad \widetilde{M}_j^s = \pi_c\Big(v_j^{(s)}\Big) \otimes \pi_{q-c}\Big(v_j^{(s)}\Big)^*, \tag{4}$$

where $N_{\text{tot}} = \sum_s N_s$ and $\text{vec}(\widetilde{M}_j^s) \in \mathbb{C}^9$. Two statistics estimate $r_{\text{eff}}$: the participation-ratio rank $r_{\text{eff}}^{\text{PR}} = (\sum_i \tilde{\lambda}_i)^2 / \sum_i \tilde{\lambda}_i^2$ and the threshold rank $r_{\text{eff}}^{1\%}$ (eigenvalues of $\mathcal{C}_c$ exceeding 1% of $\lambda_1$).

## 4.2 Results

Table 3: Formal effective-dimension measurement in $\mathrm{End}(H_{\mathrm{eff}})$, $H_{\mathrm{eff}} = \mathbb{C}^3$ (HEFF_DIM = 3). Valid/total: conjugate pairs for which `pi_c` is non-empty. $r_{\mathrm{eff}}^{\mathrm{PR}}$: participation-ratio rank. $r_{\mathrm{eff}}^{1\%}$: threshold rank. The spin-$\frac{1}{2}$ prediction in $\mathrm{End}(V_\rho)$ is $d_\rho^2 = 4$; the ambient dimension of $\mathrm{End}(\mathbb{C}^3)$ is 9.

| $q$ | valid/total | $r_{\mathrm{eff}}^{\mathrm{PR}}$ | $r_{\mathrm{eff}}^{1\%}$ | $\lambda_2/\lambda_1$ | $\lambda_3/\lambda_1$ |
|---|---|---|---|---|---|
| 29 | 5/14 | 8/3 | 3 | 1/2 | 1/2 |
| 61 | 30/30 | 8/3 | 3 | 1/2 | 1/2 |
| 101 | 50/50 | 8/3 | 3 | 1/2 | 1/2 |
| 151 | 75/75 | 8/3 | 3 | 1/2 | 1/2 |

The results are given in Table 3. For $q \in \{61, 101, 151\}$, every conjugate pair yields $r_{\mathrm{eff}}^{\mathrm{PR}} = 8/3$ and $r_{\mathrm{eff}}^{1\%} = 3$, with normalised eigenvalue structure

$$[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \tfrac{1}{2} : \tfrac{1}{2}]. \tag{5}$$

The remaining eigenvalues $\lambda_4, \ldots, \lambda_9$ are numerically zero (below the 1% threshold by a factor $> 100$). For $q = 29$, only 5/14 pairs have non-empty `pi_c` arrays; among those, the same structure (5) holds exactly. The value $r_{\mathrm{eff}}^{\mathrm{PR}} = 8/3$ is the exact algebraic consequence of (5): the normalised weights $p = [2/4, 1/4, 1/4]$ give $\sum p^2 = 3/8$, hence $r_{\mathrm{eff}}^{\mathrm{PR}} = 8/3$. Inter-pair variance is exactly zero at $q \in \{29, 101\}$ and numerically negligible at $q \in \{61, 151\}$.

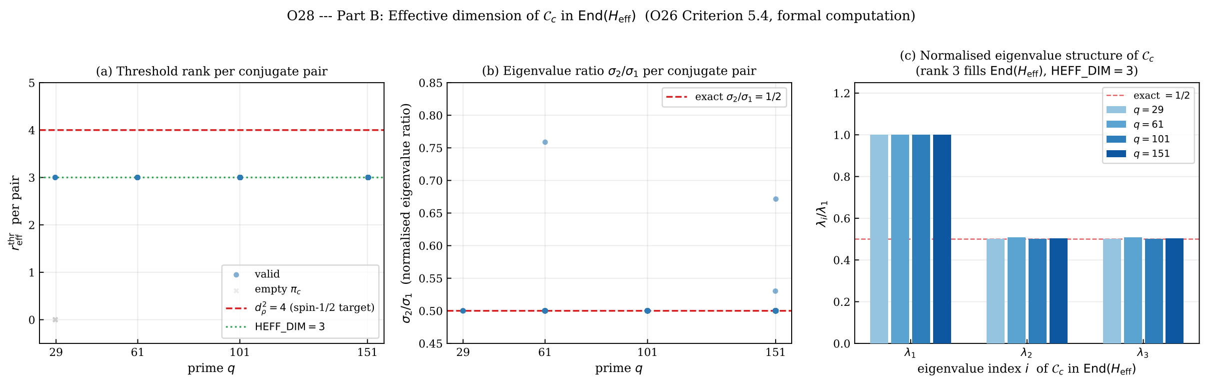


Figure 2: Part B: Effective dimension of $\mathcal{C}_c$ in $\mathrm{End}(H_{\mathrm{eff}})$, $H_{\mathrm{eff}} = \mathbb{C}^3$. (a) Threshold rank $r_{\mathrm{eff}}^{1\%}$ per conjugate pair across primes; dashed red: spin-$\frac{1}{2}$ prediction $d_\rho^2 = 4$; dotted green: HEFF_DIM = 3. (b) Normalised second eigenvalue $\lambda_2/\lambda_1$ per pair; dashed red: exact value 1/2 from (5). (c) Mean normalised eigenvalue structure $\lambda_i/\lambda_1$ for each prime; all primes give the same $[1, 1/2, 1/2, 0, \ldots]$ structure within numerical precision.

## 4.3 Interpretation

The rank-3 result fills $H_{\mathrm{eff}}$ completely: the trajectory of outer products $\widetilde{M}_j^s$ spans the full 3-dimensional eigenspace of $\mathcal{C}_c$ within $\mathrm{End}(\mathbb{C}^3)$. This is consistent with $\Sigma_c(n_3) = 3$ from O23 [10], which established that the admissible subspace at the characteristic shell depth $n_3$ has dimension 3.

The gap between $r_{\mathrm{eff}}^{1\%} = 3$ and the spin-$\frac{1}{2}$ prediction $d_\rho^2 = 4$ has a structural explanation established in O29 [8]. The anti-linear Born–Infeld parity $\rho_{q-c} = \bar{\rho}_c$ (O18) makes the conjugate-pair target $d_\rho^2 = 4$ inaccessible by construction, and $r_{\mathrm{eff}} = 3$ identifies the irreducible adjoint carrier $H_{\mathrm{eff}} \simeq \mathrm{Sym}^2(V_\rho)$ (spin-1). The spin-$\frac{1}{2}$ space $V_\rho$ ($d_\rho = 2$) is recovered as the Veronese square-root structure of the data, not as the rank measured by Test 4.

**Remark 4.1** (Resolution of O26 Criterion 5.4 in O29)**.** The result $r_{\text{eff}} = 3$ established in the present paper is explained in O29 [8] as follows. The anti-linear Born–Infeld parity $\pi_{q-c}(v) = \overline{\pi_c(v)}$ makes every outer product $\widetilde{M}_j$ complex symmetric, so the conjugate-pair target $d_\rho^2 = 4$ of O26 Criterion 5.4 is inaccessible by construction. The observed $r_{\text{eff}} = 3$ identifies the irreducible adjoint carrier $H_{\text{eff}} \simeq \text{Sym}^2(V_\rho)$ (spin-1); the spin-$\frac{1}{2}$ space $V_\rho$ ($d_\rho = 2$) is recovered as the Veronese square-root structure of the data, which lie on the cone of rank-one symmetric squares, not as the quantity measured by Test 4. The exact eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$ reflects the axial symmetry of the spin-1 carrier (the degeneracy $\lambda_2 = \lambda_3$; see Remark 4.2).

**Remark 4.2** (Structural origin of the exact degeneracy $\lambda_2 = \lambda_3$)**.** The exact equality $\lambda_2 = \lambda_3$ across all conjugate pairs and all tested primes is not a numerical coincidence. It is a structural consequence of the Born–Infeld parity involution $c \leftrightarrow q - c$ established in O18 [2], which imposes $\rho_{q-c} = \bar{\rho}_c$ on the Weil representation. This conjugation symmetry correlates the two projections $\pi_c(v)$ and $\pi_{q-c}(v)$ entering (4): the outer product $\widetilde{M}_j^s = \pi_c(v_j^{(s)}) \otimes \pi_{q-c}(v_j^{(s)})^*$ is constrained to a symmetric subspace of $\text{End}(\mathbb{C}^3)$, which forces the covariance $\mathcal{C}_c$ to have a degenerate eigenvalue structure. The degeneracy $\lambda_2 = \lambda_3$ is therefore a direct signature of the pair constraint $c \leftrightarrow q - c$ at the level of the covariance operator, and its persistence across all primes confirms that this constraint is structural, not a finite-$q$ artefact.

# 5 Discussion

## 5.1 What O28 establishes unconditionally

Three results are established from the full dataset.

First, the O14 normalisation correction is structurally coherent across the full tested range: $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, confirming that the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect.

Second, the out-of-sample extension of the window-depth measurement to $q \in \{307, 401, 503, 601\}$ excludes the linear law: the OLS fit calibrated on $q \leq 211$ overpredicts $n_1$ by +27/+49/+62/+88%, and $n_1(q)/q$ decreases monotonically from 0.138 to 0.032, so $\lim_{q\to\infty} n_1(q)/q = 0$. The asymptotic law of $n_1$ remains open; only the linear scaling hypothesis is excluded.

Third, the covariance $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ has rank exactly 3 for every conjugate pair at $q \in \{61, 101, 151\}$, with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$. This result is robust: zero inter-pair variance at $q \in \{29, 101\}$, negligible at $q \in \{61, 151\}$. The rank-3 result fills $H_{\text{eff}}$ completely and is consistent with $\Sigma_c(n_3) = 3$ from O23.

## 5.2 What remains open

The following directions are explicitly open after O28:

- The asymptotic law of $n_1(q)$: the out-of-sample campaign (now carried to $q = 601$) establishes $n_1(q)/q \to 0$ and excludes the linear law, but the sub-linear law itself is open.
- Analytical derivation of the exact eigenvalue structure (5).

The representation-theoretic direction (identification of the spin-$\frac{1}{2}$ sector $V_\rho$ and computation of $r_{\text{eff}}$ in $\text{End}(V_\rho)$) is completed in O29 [8]; see Remark 4.1 above.

# 6 Conclusion

O28 closes two open directions of the post-O27 programme. On the calibration side, $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, and the out-of-sample extension to $q = 601$ shows $n_1(q)/q$ decreasing to 0.032 and overpredicted by the linear fit by up to +88%, establishing

$n_1(q)/q \to 0$; the linear scaling hypothesis is excluded, the sub-linear law otherwise remaining open.

On the representation-theoretic side, the formal computation of O26 Criterion 5.4 in $\mathrm{End}(H_{\mathrm{eff}})$ yields a clean and robust result: the covariance $\mathcal{C}_c$ has rank exactly 3 with invariant normalised eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$ across all conjugate pairs and all primes $q \in \{61, 101, 151\}$. This fills $H_{\mathrm{eff}} = \mathbb{C}^3$ completely, consistent with $\Sigma_c(n_3) = 3$ from O23 [10]. The gap from the spin-$\frac{1}{2}$ prediction $d_\rho^2 = 4$ is resolved in O29 [8]: the anti-linear Born–Infeld parity makes $d_\rho^2 = 4$ inaccessible from conjugate-pair data, and $r_{\mathrm{eff}} = 3$ identifies the spin-1 carrier $H_{\mathrm{eff}} \simeq \mathrm{Sym}^2(V_\rho)$; the spin-$\frac{1}{2}$ space $V_\rho$ ($d_\rho = 2$) is the Veronese square-root structure.

**Data and Code Availability.** The analysis scripts `o28_analysis.py`, `o28_reff_extract.py`, and `o28_reff_figures.py`, the checkpoint files `q{q}_o25.npz`, and the summary file `o28_reff_summary.npz` are available in the o28 directory of 10.5281/zenodo.18292335.

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